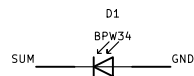
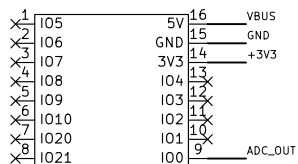
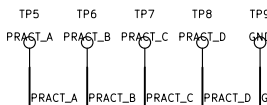


USB POWER / ESP32 ADC

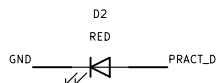
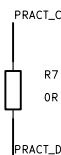
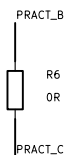
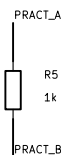
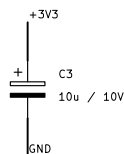
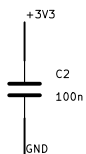
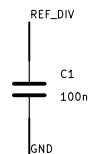
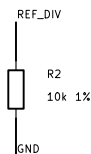
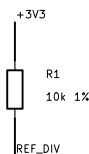
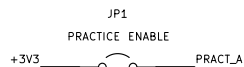
PHOTODIODE / TRANSIMPEDANCE

U1
ESP32-C3 SuperMini

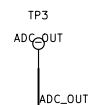
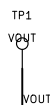
OPTIONAL SMD PRACTICE



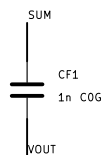
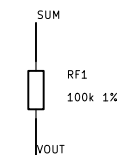
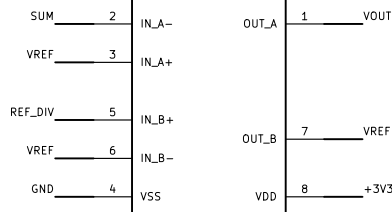
BUFFERED MIDRAIL / DECOUPLING



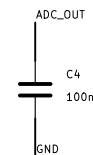
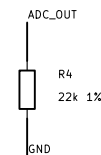
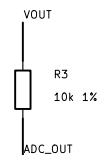
MEASUREMENT POINTS



$V_{REF} = 3.3V / 2$. $V_{OUT} = V_{REF} + I_{PH} \cdot R_{F1}$ (before clipping). BPW34 cathode = SUM, anode = GND.
RF1 is a soldered axial resistor. Power OFF before modifying feedback.
Feedback pole: $f = 1/(2 \cdot \pi \cdot R_{F1} \cdot C_{F1})$. Measure settling and clipping.
 $V_{ADC} = V_{OUT} \cdot R_4 / (R_3 + R_4)$; GPIO0, ADC1, 12dB attenuation. Calibrate ADC and dark output.
MCP6002 DIP8: A = TIA, B = VREF buffer. USB only; no external supply connected.

U2
MCP6002-I/P

ADC RANGE DIVIDER

PWR_FLAG
VBUSPWR_FLAG
GNDPWR_FLAG
GND

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File: photodiode_lab.kicad_sch

Title: PHOTODIODE TRANSIMPEDANCE LAB

Size: A4 Date: 2026-09-20

KiCad E.D.A. 10.0.1

Rev: D

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